

Positional Mapping Myanmar Text Input Scheme for Mobile Devices

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ABSTRACT

Current Myanmar PC keyboards or key mappings are very difficult to learn for novice users, and there is no efficient key mapping for mobile device keypads. In this paper, we introduce a new idea of key mapping (Positional Mapping) for phonetic scripts such as Myanmar language. Positional Mapping is a key mapping idea, based on Myanmar language characters writing positions, for small computing devices such as mobile phones, PDAs and electronic dictionaries etc. Our approach has made key mapping for Myanmar language very simple and easier to memorize. The results of first evaluation show that first time users have no problem with this key mapping and they can type Myanmar text with appropriate typing speed. Positional Mapping can be extended to other phonetic scripts such as Khmer, Thai, Indic and Lao etc.

Keywords

Text input scheme, positional mapping for small computing devices, Myanmar language, stylus-based input, soft-keyboard, user interfaces for mobile devices

1. INTRODUCTION

A number of authors have developed text input methods with various key mappings such as Multi-tap, T9 [10], MessageEase [7], FasTap, LetterWise [11] and WordWise [12] etc. for mobile devices. But most of the text input methods and key mappings are based on English, and not for phonetic based scripts such as Myanmar language. The writing systems of phonetic based scripts are based on combinations of characters and have more characters compared to English. And thus, small numeric keypads, Roman alphabet based QWERTY keyboards and soft-keyboards are not perfect for typing phonetic languages. One of the good examples is that current well known Myanmar PC keyboards use (Ctrl + Alt) key combinations to type Myanmar stacked consonants. Although there are some Romanization and transcriptions for

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Myanmar language, ordinary Myanmar people are not familiar with these transformation rules. And typing Myanmar sentences by transcription or Romanization requires more keystrokes or taps because of significant differences from Latin based scripts and writing systems. For mobile devices, keyboards or keypads have to be compact, and should support quick and easy text input. To achieve this purpose, key mapping plays a very important role. Here, we propose a new way of key mapping that is based on characteristics of Myanmar language writing system. We name it "Positional Mapping" (PM) because key mapping is logically based on Myanmar language characters writing position. By applying PM concept, we can create user friendly key mapping (i.e. all native participants in our experiments can understand key mapping within a minute) for mobile devices, which enables novice users to type Myanmar sentences easier and faster on mobile phone numeric keypads.

2. EXISTING TEXT INPUT METHODS FOR MOBILE DEVICES

In this section, we briefly discuss some of the existing text input methods or interfaces for mobile devices.

2.1 Key Based Text Input Methods

2.1.1 Multi-tap

Multi-tap is a traditional and still popular text input method for mobile phones because of its simplicity. In this input method, users press each key one or more times to get the desired letter. For example, the 3 key is pressed once for the letter 'd', twice for 'e' and three times for 'f'. In Multi-tap, users need to break (by waiting timeout or pressing timeout kill button) the key rotation sequence for typing more than one letter from the same key consecutively. One distinct drawback of Multi-tap is the keystrokes per character (KSPC) [4], which increase based on the number of letters assignment on a key. We proposed the possible Myanmar language Multi-tap keypad layout (in which consonants are ordered by alphabetically and other characters are ordered by writing order) on October 19, 2005 [8]. But we came to notice that our proposed Multi-tap keypad layout has some limitations from the follow-up analyses. They are mapping one button (i.e. Number 3 Button) for the "rarely used Myanmar consonants" and assigning many "vowel symbols, killer stroke, tone marks etc." onto only one button (i.e. Number 8 Button) [9]. We believe that key mapping or keypad layout is very important to achieve efficient tapping speed for multi-tap input methods especially for

phonetic scripts like Myanmar language. Multi-tap input method is slow (approximately $KSPC=2.03$) [4] and unambiguous.

2.1.2 Predictive Text Input (PTI)

Predictive Text Input uses an intelligent algorithm that predicts words and, in some cases, phrases and sentences while users are tapping on a keypad of a mobile phone. T9 (Text on 9 Keys) [10] is one of the famous predictive text input technology developed by Tegic Communications. T9 computes all possible combinations of a sequence of user key presses, and looks them up in the built-in dictionary. For example, to type “how”, users just need to press 4, 6 and 9 key once in order. Other famous PTI methods are iTap developed by Motorola, LetterWise by Eaton Ergonomics and eZiText by Zi Corporation. Naturally, linguistic disambiguation is not perfect for PTI methods, since multiple words may have the same key sequence. In this case, PTI gives the most common word as a default. Users can choose an alternate word by pressing next button (usually * button). The cognitive load imposed by PTI is heaviest when entering words that do not exist in the wordlist and word statistics. In this case, users have to switch Multi-tap mode and it may lead to reduce tapping performance. In general, PTI methods such as T9 and LetterWise purport to achieve performance close to $KSPC=1$ [4] [5]. There is no proper PTI method for Myanmar language yet. PTI methods for Myanmar language can be applicable for efficient text input. But the big challenge is that there is no existing usage frequency table for Myanmar characters, and digital format of lexicon and corpus etc. are not created yet. Myanmar Unicode and National Language Processing Research Center is making efforts on it [14].

2.1.3 Alternate Keypad Configuration

Alternate Keypad Configuration is also another possible way to find efficient text input method for mobile phones. Some of them are Less-Tap, MessageEase, Two-Key and SIMKEYS.

Less-Tap is the same as Multi-tap based scheme, but the key mapping is different. More frequently used characters are given precedent in the Less-Tap keypad to reduce the requirement of keystrokes. By doing this way, the typing efficiency of Less-Tap is better than Multi-tap. The character layout of the MessageEase developed by Exideas assigns unique key combinations according to the topological location of the letters on the mobile phone keypad [7]. The efficiency is improved by allocating the most frequently used characters in English with combinations. According to MessageEase keypad layout, most of the letters (except a, n, i, h, o, r, t, e, s) needs 2 taps (i.e. $KSPC=2$). Two main disadvantages of MessageEase are 1) difficult to memorize the keypad layout, and 2) not speedy because 2 taps are required to input a single letter. In Two-Key input methods, the first key selects a group of characters (e.g., the 3 key for D, E, or F), and the second key specifies the position within the group. For example, to enter the character ‘E’, users press 3 followed by 2 (E is second character in group “DEF”). Although the Two-Key input method is quite simple, $KSPC$ is also 2 for every character. Two-Key method is not very common, but it was used widely for Japanese pager Katakana text input. SIMKEYS follows an approach that is similar to how the shift key functions on a regular keyboard for assigning the rest of the characters onto the 12-key keypad. The most frequently used characters in English (i.e. A, E, I, L, N, O, R, S, T), called center plane, can be typed with only one keystroke in SIMKEYS. And the remaining characters are

separated into two groups, each containing the second most and least frequently used characters, respectively. To access the first group (i.e. B-C-D-F-G-H-M-P-U), users press the star key firstly and then the corresponding number key. Likewise, the latter group (i.e. J-K-Q-V-W-X-Y-Z) is accessed via the pound key. These two groups of letters are referred to as the star plane and pound plane [6].

In contrast, the alternate keypad configuration based on frequently used characters can improve text input speed on mobile device. On the other hand, the adequate learning time will be required by users to memorize the new keypad layout. We did one of the user studies over Myanmar keypad which is ordered by frequently used consonants. According to the experiment results, the $KSPC$ has been reduced and the performance of the text input has slightly increased comparing with alphabetically ordered Multi-tap, but text input has become difficult for novice users.

2.2 Pen Based Text Input Methods

2.2.1 Soft keyboard

A soft keyboard (sometimes called as “onscreen keyboard” or “software keyboard”) is a system that replaces the hardware keyboard on a computing device with an on-screen image map. Soft keyboards are typically used to enable input on a handheld device. Although current Myanmar PC keyboard layouts can be directly used as soft keyboard, we believe that this is not a good text input interface. This is because almost all of the current Myanmar keyboard layouts are based on traditional typewriter keyboard layout and many alphabets are hidden in the shift positions. And another problem that we noticed during our experiments is that some of the Myanmar characters are difficult to be distinguished from other similar shaped characters in small soft keyboard. For example, Myanmar characters such as (၀, ဝ, ၀, ၀, ၀, ၀, ၀, ၀), (၁, ၁, ၁) and (၂, ၂, ၂) are difficult to be distinguished.

2.2.2 Unistroke

In Unistroke [1], each letter is created with a single stroke and designed to increase accuracy of handwriting recognition techniques. Example Unistroke is “Graffiti”. In this writing system, users need to learn the new single-stroke alphabets for accuracy. If we create Unistroke characters for Myanmar characters, the user learning time will increase due to the larger alphabet size. Although this approach is simple and works well for Roman alphabets, it will not be simple for Myanmar language handwriting input. The reason is that the Myanmar script contains same shape of characters with different sizes and written in different positions.

2.2.3 Quickwriting

Quickwriting is also gesture based text input technique for continuous pen-based text entry [1]. It allows users to describe entire words or even sentences as a single continuous gesture. The text input idea of Quickwriting is to have a 3×3 grid, where characters are entered with strokes that begin in the central resting zone or home position and move through one to three adjoining positions, returning back to the resting zone. Quickwriting is about three times faster than Graffiti [3]. If we consider

Quickwriting text input interface for Myanmar language, we may need more than one grid (i.e. for consonants, dependent vowels, independent vowels, medials, finals, abbreviation characters, stack characters), and we need to shift between these grids due to Myanmar language writing system. Or another possible interface is using big grid for all Myanmar characters. Big grid is not suitable text input interface for mobile devices.

2.2.4 Cirrin (the CIRCular INput device)

Cirrin is a word-level unistroke keyboard for pen input [2]. The letters of the alphabet are arranged in a circle and ordered by usage frequency. Characters are selected by moving into and out of the appropriate segment of the circle. This tap and flick approach would be unwieldy for Myanmar language because of its larger alphabet size and writing nature.

Besides above-mentioned pen text input methods, handwriting input for Myanmar language is possible interface. But there is no proper handwriting recognizer for Myanmar language yet. And another limitation is that maximal speed is limited because users hand writing is slow.

3. MYANMAR LANGUAGE

Myanmar Script has been a majority language of Myanmar for over 1000 years old. It is the official language in Myanmar.

Consonants:

က ခ ဂ ဃ င
(ka) (kha) (ga) (gha) (nga)
စ ဆ ဇ ဈ ည
(ca) (cha) (ja) (jha) (nyya)
ဋ ဌ ဒ ဌ ဌ
(ta) (tha) (da) (dha) (na)
တ ထ ဒ ဇ န
(ta) (tha) (da) (dha) (na)
ပ ဖ ဖ ဘ ဘ
(pa) (pha) (ba) (bha) (ma)
ယ ရ လ ဝ သ
(ya) (ra) (la) (wa) (sa)
ဟ ဇ အ
(ha) (lla) (a)

Vowels:

ာ ဝ ဝ ဝ ဝ
i ii etc.

Medials:

၍ ိ ဝ etc.

Independent Vowels:

က ိ ိ ိ ိ ိ ိ

Finals:

င က ဖ ယ န etc.

Abbreviations:

၏ ၍ ၎် ၎် etc.

Punctuations:

၊ ။ etc.

Numbers:

၁ ၂ ၃ ၄ ၅ ၆ ၇ ၈ ၉ ၀

Pali Words:

အံသုမာ (နေ)၊ အဝ (သစ်ပင်)၊

ပထဝီ (မြေကြီး) etc.

Stack Characters:

ကံ ခံ ဝံ ဖံ ဖံ etc.

Figure 1. Alphabet of Myanmar Script

It belongs to the Tibeto-Burman language family and derives from Sino-Tibetan. Myanmar alphabets adapted the Mon script in the 8th century. Myanmar language has various types of characters comparing with English. (i.e. consonants, dependent vowels or

medials, independent vowels, finals and stack characters or conjunction alphabets etc.) And Myanmar language contains many Pali words especially for religious things such as praying. Despite great differences in appearance and detail, the Myanmar script follows the same basic principle as Devanagari (the Indian alphabet used to write Sanskrit, Hindi and many other languages). Overall writing direction is from left to right. In a Myanmar sentence, spaces are used to mark phrases, not dividing words.

4. MYANMAR LANGUAGE TEXT INPUT

Today we do not have standard PC keyboard layout for Myanmar language, but we have traditional type writer keyboard layout. All of the Myanmar language PC keyboards layouts are based on type writer, but key mappings and encodings are slightly different. Unicode code chart 1000-109F [16] is internationally standardized character set encoding for Myanmar script but is not widely used yet. Most of the current Myanmar keyboard layouts use many shifting (usually 3 planes such as Unshifted, Shift and Alt Planes) to fulfill the requirement of large number of keys. For example, in “WIN Myanmar” keyboard, ‘u’ key is for Myanmar character (ka) ‘က’, (Ctrl+Alt+u) for stacked character (ka) ‘ကံ’, (Shift+z) for Myanmar character (za) ‘ဇ’ and (Ctrl+Alt+Shift+z) for stacked character (za) ‘ဇံ’ etc. (Refer to WIN Myanmar keyboard layout web link [15]).

If we type according to Unicode storage order, typing Myanmar words becomes more complex and users cannot easily get accustomed to. This is because the typing order and Unicode storage order are different in Myanmar language.

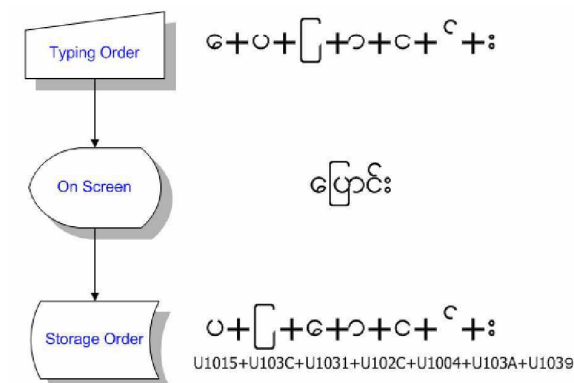


Figure 2. Typing Order and Unicode Storage Order of Myanmar Word “change”

And thus, intelligent input interface which will automatically transform from typing order to storage order is necessary. Some of the Myanmar font developers are now introducing Unicode fonts together with intelligent input interface. In practically, some old Myanmar fonts such as “WIN Myanmar font” are widely used by desk top publishing (DTP) staffs, and WIN Myanmar

5. POSITIONAL MAPPING (PM)

Positional Mapping is a concept of key mapping for mobile phone

Positional Mapping is a concept of key mapping for mobile phone based on phonetic scripts like Myanmar language writing system. When we make an analysis on Myanmar sentences, basically we can consider that Myanmar characters are written in three levels (up, normal, under) and most of the characters have their defined positions (e.g. ဧ have to be written as front vowel, ၵ, ၶ, ၷ, ၸ, ၹ, ၺ, ၻ etc. should be written as a lower character and ၾ, ၿ and ၽ etc. should be written as an upper vowel and ၼ and ၽ etc. should be written right side of the consonants). But there are several exceptional cases, e.g. although consonant က (ka) is usually written in the normal level, when it becomes 'subscripted ka' or 'stack character' it becomes smaller than normal ka and written in the lower level as 'က' in 'ပကက'.

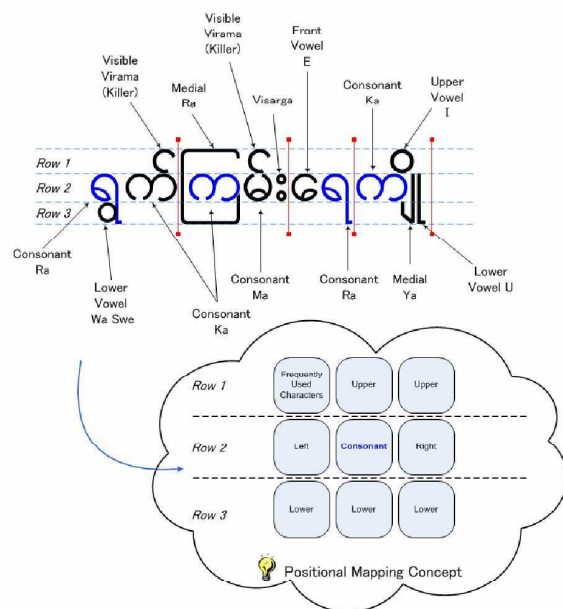


Figure 3. Positional Mapping for Myanmar Mobile Phone

We divide mobile phone keypad into three levels, e.g. 1, 2 and 3 keys are upper, 4, 5 and 6 keys are normal or center and 7, 8 and 9 keys are under. In the normal level, 4 key is seen as the front part, 5 key as the center position and 6 key as the rear part. According to the Myanmar language writing method, we assigned upper vowels and killer at 2 and 3 keys, front vowels on 4, main consonant on 5 key, rear vowels on 6 keys and lower vowels and medials on 7, 8 and 9 keys. Here, the consonants will be shown as

a list. We assigned “frequently used Myanmar characters and words” such as ‘ဝါ’, ‘လား’, ‘၏’, ‘ထို’, ‘ထို’, ‘ဝါ’, ‘သူ’, ‘ကျေ’ and ‘ဒီ’ etc. at number 1 key. This mapping concept is simple, and its keypad layout is very easy to memorize for users who understand the Myanmar language writing system. Although PM concept is based on key mapping of mobile phones, it can be applied not only for mobile phone but also for other mobile devices such as PDA and Tablet PC etc.

6. PROTOTYPE IMPLEMENTATION

We have implemented PM input scheme for the windows desktop and PDA. For the windows desktop, we implemented two input user interfaces; 1) User interface for experiment with numeric keypad and 2) User interface for mobile phone text input. Here, we present our three prototypes, which are all developed with Microsoft Visual Studio .Net 2003.

6.1 Prototype for Mobile Phone

This is the very first prototype that we created for simulation environment of PM text input for mobile phones. With this prototype, key mapping can be changed through configuration file. Configuration file is the text file format that contains the key assignments for each button.

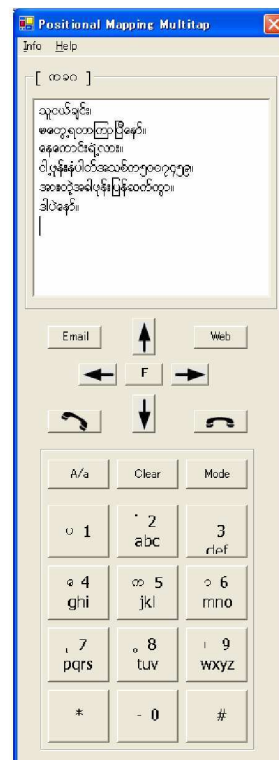


Figure 4. PM Prototype for Mobile Phone

[For Myanmar Characters]

၁ = ဝါ,တယ်လား,လဲ,သည်,၏,ထို့,ဖြင့်,ဟို,ဒီ့အံ့ဝါ,
နင်,သူ့မင်း,ညည်းရှင်,ရှင်သင်ကျပ်,ကွန်တော်,
ကွန်မ,ကျနော်,ကျမ,ကွန်ပုံ,ကျပ်ပြာ,ခွဲ,နာရီ,မိနစ်,
ရာဇာဏိုင်,သောင်း,သိန်း,သန်း,ကုမ္ပဏီ etc.

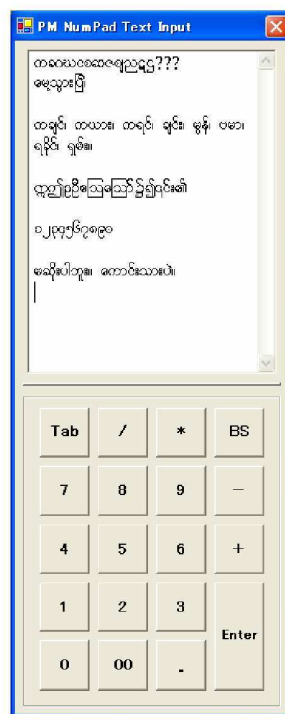
၂ = ၊
၃ = ,
၄ = -
၅ = က,ခ,ဂ,ဗ,င,စ,ဆ,ဇ,ဈ,ည တဲ့ အ
၆ = ပေး၊-း
၇ = မှတ်ချက်
၈ = စီး,ဆုံး,
၉ = space,||
၀ = ကေ,ခေ,ဂေ,ယူ,ဖေ,ဆေ etc.
* = ကျ,ဤ,ဥ,ဓ,ဩ,ဍု,ခြ်,ဝင်း,ကြော့်,ဦးနှု,သာ,ညှ, & etc.
= @,,.,,,,_,-\$,%!,?,&, #, +, *, =, /, (,), <, >, [,], { } etc.

[For Myanmar Numbers]

၁ = ၁
၂ = ၂
၃ = ၃
၄ = ၄
၅ = ၅
၆ = ၆
၇ = ၇

6.2 Prototype for Numeric Keypad

keypad (although program works with software buttons, we do not allow users to use during user experiments), 2) users cannot see the key assignment on the software buttons and 3) users have no chance to use arrow keys to choose a character like in mobile phone prototype. We created these differences in order to know how user tapping speed changes and users respond. In this prototype, users need to press “Enter” to type a character.



6.3 Prototype for PDA

The third PM prototype is for Personal Digital Assistant (PDA). We created this prototype in order to prove that our PM can be applied not only for key based text input methods but also for pen based text inputs. With PDA, users can easily type Myanmar language with stylus. Typing speed will go up compared to previous two prototypes, since users can directly go to the desire letters shown on the PDA screen. The logic of the letter assignment on PDA is almost the same as previous two prototypes. An exception is that we put all of the lower medials and vowels all together on a button (i.e. ‘ $\bar{1}$ ’ Button) (Please see Figure 5 and Figure 7). In the above two prototypes, we divided these lower medials and vowels into three groups and assigned onto three buttons (i.e. 7, 8, 9 buttons in mobile phone prototype, and 1, 2, 3 buttons in numeric keypad prototype) to reduce the requirement of keystrokes. The keystrokes or taps requirement to type a Myanmar character with PDA is two. The drawback of this prototype is that users have to go back to home menu screen (See

Figure 7) after typing a character or some characters (i.e. in case of medial characters).

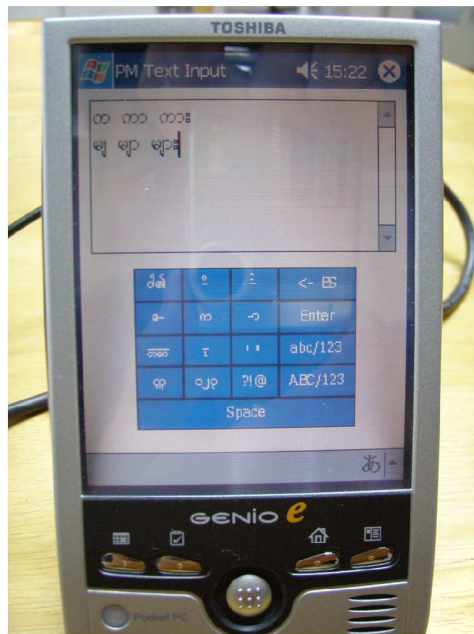


Figure 7. PM Prototype for PDA

7. CHARACTER COMBINATION ERROR CHECKING

The important function in PM Myanmar text input method is “character combinations error checking”. Users can make impossible combinations of Myanmar characters when they type. The System should support error checking for impossible combinations. This is like spell-check feature in English. In Myanmar language, however, we need to take two steps of checking; first is checking of impossible or meaningless combinations (character level), and the second is checking of word level spelling mistakes. In our program, we support only characters combinations error checking. There is no proper spelling checking engine for Myanmar language yet, and we assume that this is not necessary for small computing devices. An example of meaningless Myanmar characters combinations are “ကု”, “နီ”, “ဝေး”, “နီ” etc.

8. USER STUDY WITH 3 PROTOTYPES

To evaluate the performance of “Positional Mapping”, we made a user study with five native participants who are between 23 and 31 years old. All of the participants are familiar with PC but they don’t have an experience of text input with mobile phone and numeric keypad. All are touch typists in English (with QWERTY

keyboard) but not in Myanmar language. We used our developed three simulation programs (mobile phone, numeric keypad, PDA) for user experiments. No participants had any prior experience with our text input simulation prototypes. The PDA used for experiments is Toshiba GENIO e550G (ARM PXA250 processor, 400 MHz) running Microsoft Pocket PC3.00 OS with 64MB RAM. And we used ELECOM numeric keypad for numeric keypad text input prototype. The experiment procedure is as follows:

1. Explaining the concept of PM
2. Explaining the key assignment for all prototypes
3. Making demonstration of text input with each model
4. Allowing 20 minutes practice time for each user to learn text input methods with prototypes (4 multi-tap models of mobile phone prototype, numeric keypad prototype and PDA prototype)
5. Making user experiment
6. Discussing with users and learning users’ responds

We recorded the total time spent to type 6 sentences Myanmar SMS message (including error correction time) for 10 times with each model. We also made analysis on keystroke requirement of each prototype for Myanmar consonants “Ka to A”, Myanmar vowels “A, Ar, I, II etc.”, some frequently used Myanmar words, short Myanmar SMS message and some of the selected stack words (see Figure 8).

(a)	(b)
အအာကုဏ္ဍိညဉ္ဇအသြဇေဝိ အံအား	ဝါတယ်လားလဲသည်၏ထိုမြင့်
အအာအိအီအုအူအေအိအောအော်အံအား	ဟိုဒီအဲဒါနင်သူမင်းသည်းရှင်
ကကကကကကကကကကကကကကကကကက	ရှင်သင်ကျပ်ကျွန်တော်ကျွန်မ
(c)	ကျနော်ကျမကျပ်ပြားခွဲနာရီ
သူငယ်ချင်း၊	မိနစ်ရာတောင်သောင်းသိန်း
မတွေ့ရတာကြာပြီနော်။	သန်းကုဋေ
နေကောင်းရဲ့လား။	(d)
ငါ့ဖုန်းနံပါတ်အသစ်က၇၀၈၇၄၅၉။	တက္ကသိုလ်၊ဝဏ္ဏ၊ဥက္ကာ၊ဥက္ကဋ္ဌ၊ပိန္နာ၊
အားတဲ့အခါဖုန်းပြန်ဆက်ကွာ။	သိပ္ပံ၊ပြဟာလက္ခဏာ၊ဝေသာန္တရာ၊
ခါပဲနော်။	ယောက္ခမ။

Figure 8. Myanmar text for user study (a) Myanmar vowels A, Ar, I, II etc. (b) some frequently used Myanmar words (c) short Myanmar SMS message (d) Myanmar stacked words

9. EVALUATIONS

In this section, we discuss the results of user experiments. First of all we discuss for mobile phone prototype, and then for numeric keypad and PDA.

When we compare keystrokes requirement of four different key mappings for mobile phone, we found that Positional Mapping

requires more keystrokes than other three prototypes (see Figure 9). The reason is that all of the 33 consonants are assigned as a list only on No.5 key. And thus, the typing consonants require some arrow keys movement. But this model enables the highest tapping speed because users can type consonants easily by selecting from the list. Another advantage of this model is its user friendliness. All of the participants chose “Positional Mapping” or “Positional Mapping with Myanmar consonants ordered by frequency” keypad layout as the best input interface. We believe that the typing speed can increase if we combine PM with intelligent predictive method for characters or Myanmar words in the future. In this paper, we made all of the user experiments for mobile phone using with simulation program and computer mouse. User tapping speed can go up with real mobile phones. We are now preparing to make user study with real mobile phone. From the results of user experiments, it is proved that Positional Mapping is easier to memorize and can achieve appropriate tapping speed (around 22.5% faster than proposed Multi-tap). Here, we are interested only in maximum tapping speed because we would like to know how novice users can type faster with Positional Mapping keypad layout. The maximum typing speed of PM is 4 min 16 sec, and the one of PM with consonants ordered by usage frequency is 4 min 00 sec. The maximum tapping speed of four different key mappings is shown in Table 1.

Table 1. The maximum tapping speed (min:sec) for 4 Multi-tap prototypes by 5 native users

Input Method	User1	User2	User3	User4	User5
Multi-tap	6:52	9:33	6:15	5:15	4:26
Multi-tap (Frequency)	5:42	6:56	5:58	4:47	5:13
Multi-tap (PM)	4:29	4:43	4:16	4:24	4:20
Multi-tap (PM Frequency)	4:17	4:00	4:18	4:03	4:04

PM keypad layout experiments with numeric keypad give maximum typing speed 4 min and 29 sec to finish typing 6 sentences of Myanmar SMS message (see Figure 8). The difficulty of this prototype that we found from user experiments was that most of the users were not familiar with numeric keypad. They only use one finger (middle finger or index finger) to type SMS message. Most of the users put numeric keypad on the table while typing, and only one user holds the numeric keypad in his hand and types. One good point that we noticed is that user typing speed gradually increased during 10 experiments time. Overall typing speed of this prototype is slower than mobile phone prototype or PDA prototype. Although most of the users had slower typing speed with this prototype, one user (User2) achieved his fastest typing speed with numeric keypad prototype.

Table 2. The maximum tapping speed (min:sec) for 3 Positional Mapping prototypes by 5 native users

Input Method	User1	User2	User3	User4	User5
Mobile Phone	4:29	4:43	4:16	4:24	4:20
Numeric Keypad	6:07	4:29	5:10	4:52	5:21
PDA	4:05	5:00	4:10	3:58	4:01

With PDA device, most of the users can type faster than the other two prototypes. The maximum typing speed reaches 3 min and 58 sec. All of the users hold PDA in left hand and stylus in right hand. Our PM PDA prototype enables appropriate Myanmar text entry speed and very easy typing. However, some of the Myanmar characters (especially medials and stacked characters) cannot be shown properly on the PDA screen in this model. Some medials can appear only half portion of its character glyph. This may be because of the Myanmar font that we used.

As a result of the follow up discussions with participants, we received valuable suggestions and responds. Some of them wanted slang words to be added to frequently used Myanmar characters groups, and also some useful symbols for Myanmar people to be added such as “symbol for pagoda”, “symbol for monastery” and “symbol for Mandalay city” etc. All of them responded that they had no difficulty in memorizing PM keypad layout. One of them mentioned that he was very fond of order of medials like “၂”, “၃” and “*” because it is useful in typing the word like “၂၃*”. One of them wanted key or function with prototypes to be deleted because our prototypes support only backspace function.

10. CONCLUSIONS

We have proposed a new key mapping scheme for Myanmar language based on the Myanmar language writing system, and the mapping is compact and easy to memorize. The results of user experiments prove that it is applicable not only for mobile phones but also for pen based mobile devices such as PDA. This also seems to be a promising approach to other phonetic scripts such as Khmer, Thai, Indic and Lao etc. We also plan to develop PM prototypes for other Myanmar ethnic groups’ languages such as Kayin, Shan and Mon etc. In this paper, user experiments for mobile phone were made by using PM simulation program, and we are planning to conduct user study with real mobile phones in the near future.

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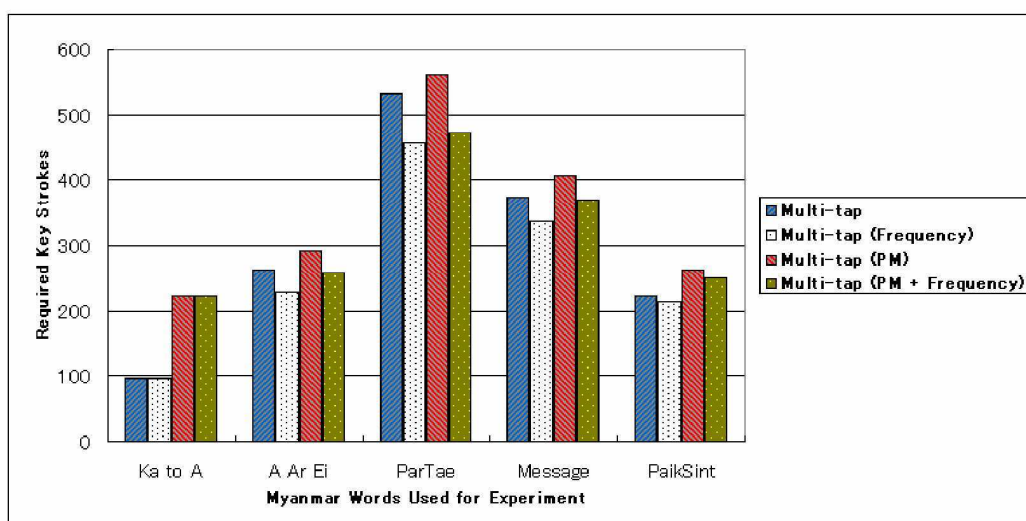


Figure 9. Keystrokes comparison for four Multi-tap mobile prototypes (here X axis Myanmar text used for user experiments and Y axis represents required keystrokes to finished)

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